TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-based Optimization of Production Processes and Predictive Maintenance			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI √), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	ICT			
	Medium Category	Small Category	Medium Category	Small Category
	Digital content	Digital contents production and distribution		
Concept of Proposed Technology	AI based analysis of existing data; Fusion of data of different origin and kinds; Transfer to other production processes; Optimization of quality assurance, repair, and maintenance.			
Rationale for Joint R&D Project	International cooperation is necessary for fields aimed at solving global challenges such as climate change. Digitalization and AI in production processes are current global and Korean trends.			
Key Words	1. AI-based Analysis 2. Predictive Maintenance 3. Process Optimization 4. Fault Detection 5. Data Fusion			

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	Optimized production process, predictive fault and failure detection.			
R&D Target Specifications	• Reduced production costs and energy consumption • Improved wear-related maintenance intervals			
R&D Plan	Timeline: 3 Years. Plan: Development and usage of AI methods for evaluation of data. Focus on the fusion of data of different origins and kinds. Transfer findings to other production processes. Execute optimization of quality assurance, repair, and maintenance.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Development and usage of AI methods for evaluation of data • Applying more than 20 years of experience in machine learning and AI technologies • Analysis of various technical, medical, and bio signals • Development of secure and operationally reliable AI systems • Handling and processing of large and small amounts of data
Expected R&R of Korean SME Partner	The Korean partner is expected to provide both the production process and the data acquisition.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Outcomes: Establishing cooperation with Korea and building on this to acquire further joint projects. Impact: Reduction of production costs and energy consumption through optimized processes and predictive fault detection.

6. References

(1)

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization